

Carbon market pilot policies and corporate green governance performance: Evidence from China

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Abstract

This study employs a multi-period difference-in-differences model to analyze 2009–2022 corporate data, assessing the impact of China's carbon market pilot policy on corporate green governance performance, including environmental disclosure, green innovation, and stakeholder engagement. The policy enhances green governance through two channels by easing corporate financing constraints and stimulating green technology innovation, with results robust to propensity score matching difference-in-differences and spatial placebo tests. Notably, effects are stronger for non-state-owned enterprises, central-region firms, and firms led by environmentally proactive executives. We propose tiered carbon pricing and targeted green financing to address regional and ownership-based disparities, guiding strengthening of market-based green governance in emerging economies.

KEYWORDS

carbon market pilot policy, corporate green governance performance, heterogeneity analysis, mediation effect, multi-period difference-in-differences

JEL CLASSIFICATION

Q58, Q56, G38

1 | INTRODUCTION

The need to balance economic growth and carbon neutrality globally has made carbon emissions trading systems (ETS) a key climate policy focus. As one of the world's largest emitters, China's carbon market pilot policies, initiated in 2013, represent a pivotal experiment in market-based corporate green governance. While existing studies primarily focus on macroeconomic outcomes, including emission reductions (Zhang et al., 2017) and gains in green productivity (Yang et al., 2021), the micro-level implications for corporate green governance remain underexplored. Corporate green governance refers to the ways in which firms strategically integrate environmental accountability, green innovation,

Abbreviations: APPCAP, Air Pollution Prevention and Control Action Plan; CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; CSR, corporate social responsibility; ESG, environmental, social, and governance; ETS, Emissions Trading System; GDP, gross domestic product; GGP, green governance performance; GTFP, green total factor productivity; IoT, Internet of Things; IPC, International Patent Classification; IPO, initial public offering; IV, instrumental variable; LCP, Low-Carbon Pilot; MD&A, management discussion and analysis; NLP, natural language processing; non-SOEs, non-state-owned enterprises; OLS, ordinary least squares; PSM, propensity score matching; PSM-DID, propensity score matching difference-in-differences; ROE, return on equity; SA, size-age index; SOEs, state-owned enterprises; TF-IDF, term frequency-inverse document frequency.

and stakeholder engagement into organizational decision-making. This concept reflects a shift from regulatory compliance toward proactive and embedded environmental management, which is an essential step toward systemic decarbonization. However, it remains unclear whether carbon pricing mechanisms can move beyond compliance to foster proactive governance transformations, particularly within the complex institutional context of emerging economies. This gap calls for rigorous empirical analysis.

Prior research has examined the macroeconomic impacts of carbon markets, but evidence on firm-level governance responses remains fragmented. For instance, early evaluations of the EU ETS emphasize volatility and weak compliance incentives, limiting structural transformation (Streimikiene & Roos, 2009). In China, regional pilot policies have led to spatially heterogeneous outcomes: while Guangzhou saw marked carbon intensity reductions (Cheng et al., 2015), other regions experienced limited gains in green total factor productivity (TFP), partly due to policy design asymmetries. Mechanistically, recent work explores transmission channels such as industrial relocation (Ren & Liu, 2024), geographic spillovers in technology adoption (Li & Wang, 2022), and the moderating role of political incentives (Zhu & Zhang, 2016). These studies highlight the contextual complexity of carbon market effects, but they largely focus on productivity or emissions outcomes, rather than internal shifts in corporate governance structures. Huang et al. (2022) further note a paradox: while carbon markets stimulated green innovation in high-emission sectors, they simultaneously curtailed government-led green investments, exposing tensions between market and fiscal instruments. Despite these contributions, existing research remains largely focused on productivity or emissions outcomes and fails to capture the internal governance restructuring induced by carbon markets.

This narrow scope obscures deeper organizational dynamics critical to understanding sustainable transitions at the firm level. Three critical gaps persist. First, existing literature conflates green innovation with holistic governance performance, overlooking structural dimensions such as board oversight and supply chain decarbonization. Second, the dual pathways through which carbon pricing operates, namely market incentives and institutional coercion, are rarely disentangled. Third, the temporal dynamics of policy effects, particularly post-2020 amid China's national carbon market expansion, have not been adequately addressed.

This study bridges these gaps by examining how China's carbon market pilots reshaped corporate green governance performance (GGP). Using a multi-period difference-in-differences (DID) design, we analyze a proprietary dataset covering the period from 2009 to 2022 and encompassing 3500 listed firms in both pilot regions (i.e., Beijing, Tianjin, Shanghai, Chongqing, Hubei, Guangdong, and Shenzhen) and non-pilot regions. Our approach diverges from prior research in three key areas. First, we develop a comprehensive measure of corporate GGP that captures firms' integration of environmental priorities into strategic and operational decision-making. This measure is constructed as a composite index of three dimensions: environmental disclosure quality (via natural language processing [NLP] analysis), green innovation (proxied by patent intensity), and stakeholder engagement (drawn from environmental, social, and governance [ESG] ratings and corporate social responsibility [CSR] reports), which together reflect the informational, technological, and organizational pillars of governance. Second, we investigate the mechanisms through which carbon pricing influences corporate governance by explicitly examining two distinct channels: the alleviation of financing constraints for green investments (Hering & Poncet, 2014) and regulatory pressure induced by penalties (Fredriksson & Millimet, 2002). These mechanisms are formally evaluated within a mediation framework. Finally, we analyze dynamic heterogeneity in policy effects by conducting subgroup DID analyses across firm ownership (state-owned enterprises [SOEs] vs. non-SOEs), regional development tiers, and levels of executive green cognition. Although we do not employ formal causal machine learning estimators, we leverage machine-learning-based stratifications, including ESG-driven clustering and textual profiling, to capture potential non-linear heterogeneity in treatment effects (Li & Zhou, 2005).

Our findings reveal that the carbon market pilot policy enhances corporate GGP by an average of 12.7%. This effect is particularly pronounced among non-SOEs, firms in central regions, and those led by environmentally conscious executives. Robustness checks, including instrumental variable approaches to address endogeneity and spatial placebo tests, confirm the validity of these results. Notably, the policy's impact appears to evolve over time. After 2020, as the national carbon market expanded, firms began to move beyond short-term compliance, showing early signs of adapting their governance structures to align with long-term environmental objectives, evidenced by a 23% increase in board-level environmental committees. While these developments are encouraging, they reflect incremental adaptation rather than wholesale transformation, highlighting the ongoing nature of governance change in response to carbon pricing.

This research contributes to three key strands of literature. Theoretically, it synthesizes institutional and capability theories, proposing a "coercive-enabling" framework that challenges the traditional state-market dichotomy in

corporate GGP. Methodologically, it introduces NLP-based textual analysis to quantify qualitative governance disclosures, thereby reducing biases inherent in manual coding. From a policy perspective, it advocates tiered carbon pricing aligned with regional industrialization stages and complementary green financing support for non-SOEs, with implications for global ETS design.

2 | THEORETICAL FOUNDATIONS AND ASSUMPTIONS

Based on the theoretical framework, we develop a set of testable hypotheses (H1–H3).

2.1 | Direct impact of carbon market pilots on corporate green governance

The implementation of the carbon market pilot policy provides empirical support for the Porter Hypothesis by internalizing emission costs and incentivizing firms to innovate (Porter & van der Linde, 1995). This hypothesis posits that well-designed environmental regulations can drive technological innovation and enhance operational efficiency, transforming compliance costs into competitive advantages. Evidence from China's pilot programs aligns with this mechanism. For instance, firms in Shanghai reduced carbon intensity by 14.7% through energy-efficient retrofits and process optimizations aimed at minimizing carbon-related costs (Li et al., 2024). Similarly, Guangdong's pilot policy led to a 19% decline in regional carbon intensity, as firms adopted cleaner production methods to meet emission quotas (Cheng et al., 2015). These outcomes underscore the role of carbon pricing in integrating environmental considerations into corporate decision-making, reinforcing the potential of market-based policies to drive sustainable industrial transformation.

Institutional theory further provides a valuable framework for understanding the impact of carbon market policies by highlighting how regulatory pressures reshape corporate norms and governance structures (Scott, 2014). By establishing a clear compliance framework that includes emission quotas, trading mechanisms, and penalties for non-compliance, carbon markets compel firms to formalize environmental accountability. Empirical evidence supports this institutional effect. Post-2020 data reveals a 23% increase in board-level environmental committees among pilot firms, reflecting structured oversight of sustainability goals in response to institutional demands (Yang et al., 2021). This form of institutional coercion aligns with findings from the EU ETS, where stringent monitoring and enforcement mechanisms played a critical role in achieving emission reductions (Ellerman & Buchner, 2008).

Dynamic capability theory complements these perspectives by underscoring firms' adaptive responses to regulatory shocks (Teece, 2014). Carbon pricing incentivizes resource reallocation toward green innovation and stakeholder engagement. Jing et al. (2024) found that emission trading enhanced green TFP primarily through efficiency gains, such as reductions in energy waste, rather than radical technological breakthroughs. This outcome aligns with the “induced innovation” hypothesis, which suggests that firms prioritize incremental improvements to meet regulatory demands while minimizing operational disruptions (Aghion et al., 2016). For instance, pilot firms increased green patent filings by 23%, with a focus on renewable energy (IPC-Y02E) and pollution control technologies (IPC-Y02C), improving governance metrics such as environmental disclosure quality (Huang et al., 2022).

However, the policy's efficacy hinges on credible enforcement and equitable quota allocation. Early phases of the EU ETS were undermined by overallocation, which weakened innovation incentives (Ellerman & Buchner, 2008). In contrast, China's pilot regions implemented dynamically adjusted quotas and stringent penalties, fostering stronger governance improvements. Non-state-owned enterprises (non-SOEs) in central China, facing more rigorous compliance checks, reduced financing constraints by 18% through green bond issuances, facilitating long-term sustainability investments (Liu et al., 2021).

H1 The carbon market pilot policy is positively associated with corporate green governance performance.

2.2 | Mediating role of financing constraints

The policy serves as a regulatory signal (Spence, 1973), enhancing transparency regarding firms' environmental risks and compliance capabilities. In pilot regions such as Guangdong and Shanghai, financial institutions have developed

specialized green credit assessment tools incorporating carbon performance metrics. For example, banks in Guangdong introduced preferential loan terms for firms meeting emission reduction targets, lowering borrowing costs for low-carbon projects by 18% (Huang et al., 2022). Similarly, in Shanghai, loan approvals increased by 15% for firms adhering to carbon disclosure mandates (Cheng et al., 2015). These measures illustrate how the policy translates regulatory requirements into financial incentives, facilitating firms' access to capital for energy-efficient upgrades.

The alleviation of financing constraints is empirically captured by the size-age (SA) index, a widely used measure of financial frictions (Hadlock & Pierce, 2010). Firms in pilot regions experienced a statistically significant reduction in SA index values ($\beta = -0.030$, $p < .01$), indicating improved access to external capital (Liu et al., 2021). This facilitated resource reallocation toward green initiatives, such as renewable energy adoption (e.g., solar panel installations) and supply chain decarbonization (e.g., supplier sustainability audits). This effect may be particularly relevant for non-state-owned enterprises (non-SOEs), which typically face higher financing barriers due to limited state support, suggesting that the policy may help alleviate credit market disparities (Liu et al., 2021).

Institutional investor behavior further reinforces this mediation mechanism. Following policy implementation, ESG-focused institutional ownership in pilot regions increased by 34%, reflecting heightened investor demand for climate-aligned assets (Dyck et al., 2019). In response, firms strengthened governance practices, such as real-time emissions monitoring and third-party environmental audits, to attract ESG-conscious capital. In Guangdong's pilot zone, green bond issuances rose by 25%, driven by investor preferences for firms with transparent carbon management strategies (Huang et al., 2022).

The policy's effectiveness varies across ownership structures and regional context. Non-SOEs, which rely on external financing, leveraged green credit incentives to secure low-cost capital for long-term sustainability projects. In contrast, SOEs, which benefit from preferential access to state-backed loans, exhibited weaker responses (Liu et al., 2021). Regional disparities emerged: provinces with well-developed financial markets (e.g., Guangdong) experienced stronger mediation effects due to investor confidence in policy enforcement, whereas less developed regions faced challenges due to fragmented regulatory oversight. To address these disparities, complementary measures are essential, including green credit guarantees and standardized carbon disclosure. For example, in 2020, Hubei province introduced a carbon-linked loan guarantee program, reducing collateral requirements for low-carbon projects by 30% and accelerating capital deployment (Liu et al., 2021).

H2a The carbon market pilot policy enhances green governance performance by alleviating financing constraints.

2.3 | Mediating role of green technology innovation

The carbon market pilot policy stimulates green technology innovation through a dual mechanism: it penalizes non-compliance while rewarding firms that adopt low-carbon technologies (Jaffe et al., 2002). This aligns with resource-based view, where firms leverage innovation to transform regulatory pressures into market competitiveness (Barney, 1991). Empirical evidence from China's pilot regions reveals that regulated firms increased green patent applications by 23%, particularly in renewable energy (e.g., solar panel efficiency) and emission control technologies (e.g., carbon capture systems), as they sought to reduce compliance costs and capitalize on emerging green markets (Li et al., 2024). For instance, industrial firms in Shanghai lowered carbon emissions per unit of output by 14.7% through process innovations such as AI-driven energy management systems and waste heat recycling (Huang et al., 2022). Media scrutiny reinforced this trend: firms with negative environmental exposure increased R&D investments by 34% compared to peers, aiming to mitigate reputational risks and align with stakeholder expectations (Aerts & Cormier, 2009). These innovations also strengthened corporate green governance, including improved environmental disclosure and supply chain decarbonization.

The policy's impact on innovation is moderated by ownership structure and sectoral dynamics. Non-SOEs, constrained by market-driven financing, exhibited stronger innovation responses. In Jiangsu Province, non-SOEs allocated 25% of carbon trading revenues to green R&D, filing 30% more patents than SOEs (Liu et al., 2021). In energy-intensive sectors such as steel and cement, firms prioritized incremental innovations (e.g., fuel-switching), reducing emissions by 18%. In contrast, technology firms prioritized digital solutions such as smart grids and blockchain-enabled carbon tracking (Huang et al., 2022). Regional disparities also emerged: Guangdong, with its mature financial markets and a price floor of ¥40/ton (approx. €5.30/ton), recorded 28% higher patent filings than regions with volatile carbon prices, highlighting the role of stable policy design (Cheng et al., 2015). In contrast, abrupt quota cuts in the early EU ETS

phases stifled innovation. This experience later informed China's phased approach to quota adjustments (Ellerman & Buchner, 2008).

H2b The carbon market pilot policy improves green governance performance by stimulating green technology innovation.

2.4 | Policy design and market risk moderation

The effectiveness of the carbon market pilot policy in enhancing GGP is largely shaped by institutional design and risk management strategies. Early lessons from the EU ETS show that over-allocated quotas caused carbon prices to collapse below €5/tCO₂e, which in turn undermined innovation incentives (Ellerman & Buchner, 2008). These experiences highlight the importance of adaptive mechanisms. China's pilot regions addressed these challenges by implementing dynamic quota adjustments and hybrid regulatory–market tools. For example, Guangdong Province reduced quotas by 12% annually from 2014 to 2022, stabilizing carbon prices between ¥40–¥60/tCO₂e, which incentivized firms to adopt emission-reducing technologies such as waste heat recovery systems (Cui et al., 2018). This dynamic approach contrasts with the EU's initial static model, illustrating how iterative quota revisions can balance market stability with environmental objectives.

Robust monitoring systems and complementary policies further amplify policy efficacy. For example, Beijing's pilot program incorporated IoT (Internet of Things)-based emission tracking, achieving 89% compliance rates through real-time data verification, which represents a clear improvement over traditional self-reporting practices (Huang et al., 2022). Regions that combined strict oversight with price stabilization tools, such as Guangdong's ¥40 per ton price floor, observed 34% stronger corporate GGP than those relying solely on market mechanisms. Shenzhen also introduced additional safeguards, including border carbon adjustments, which imposed tariffs on imports from regions with weaker climate policies, mitigating carbon leakage risks (Böhringer et al., 2012). These measures reduced cross-regional arbitrage and ensured equitable competition, fostering long-term investments in green technologies.

Policy outcomes vary depending on regional institutional capacity. Mature regions, like Guangdong and Beijing, allocated carbon revenues (e.g., ¥2.3 billion over 2015–2022) to fund industrial energy efficiency upgrades, driving a 19% reduction in carbon intensity (Cheng et al., 2015). In contrast, regions with fragmented enforcement, such as early-phase Hubei, faced price volatility and non-compliance issues. These disparities highlight the need for centralized monitoring, fiscal coordination, and phased implementation. The success of China's pilot programs aligns with institutional theory, which emphasizes the importance of enforceable rules and adaptive governance (Scott, 2014). To optimize carbon markets, emerging economies should prioritize dynamic quotas, price stability mechanisms, and complementary regulations like BCAs, ensuring both environmental rigor and market confidence.

H3 The policy's impact on green governance performance is amplified in regions with robust regulatory frameworks and risk mitigation mechanisms.

3 | RESEARCH DESIGN: MODEL SPECIFICATION, VARIABLES AND DATA

3.1 | Sample selection and data sources

This study utilizes a panel dataset of Chinese A-share listed firms spanning 2009–2022. Financial and governance variables, including firm size, leverage, and board characteristics, were sourced from the Wind Economic Database and the China Stock Market & Accounting Research (CSMAR) platform. Green innovation metrics, measured by low-carbon patent applications, were obtained from the China National Intellectual Property Administration (CNIPA) and filtered using International Patent Classification (IPC) codes related to renewable energy and pollution control technologies. Carbon market policy details, including pilot region quotas and transaction records, were retrieved from the National Carbon Emissions Trading Registry and cross-verified with provincial environmental bureau announcements. Regional economic indicators, including gross domestic product (GDP) and industrial structure, were derived from the National Bureau of Statistics and the China Environmental Statistical Yearbook.

The initial sample included 6820 non-financial firms. Financial institutions (128 firms) were excluded due to divergent accounting standards, and firms under special treatment (ST or *ST, 472 firms) were removed to mitigate biases associated with financial distress. Observations missing more than 20% of critical variables (e.g., carbon emissions, R&D expenditure) were excluded. Continuous variables were winsorized at the 1st and 99th percentiles to address outliers. The final data comprises 20,727 firm-year observations, covering all eight pilot regions. To ensure consistency across sources, fiscal year-end dates were aligned with calendar years and discrepancies in firm identifiers were reconciled across databases. The baseline DID regressions are estimated using the full sample, while the propensity score matching difference-in-differences (PSM-DID) analysis relies on the matched subsample described below.

3.2 | Model construction

Benchmark regression model: To examine the direct impact of the carbon market pilot policy on corporate GGP, we construct a difference-in-differences model (DID):

$$GGP_{it}^k = \alpha^k + \beta^k did_{it} + \sum_{j=1}^n \gamma_j^k X_{jit} + u_i^k + \lambda_t^k + \varepsilon_{it}^k. \quad (1)$$

In the model, i denotes the firm, t represents the year, and k represents the specific dimension of corporate GGP (or the composite index). The dependent variable, GGP_{it} measures the GGP of firm i in period t ; the key independent variable, did_{it} is a dummy indicating exposure to the carbon market pilot policy. If the firm is located in a pilot province or city after policy implementation, $did_{it} = 1$, otherwise $did_{it} = 0$; α is a constant term; the coefficient β captures the policy's effect on GGP; a significantly positive estimate supports H1.

The model includes a vector of control variables, X_{it} , accounting for firm and regional characteristics that may affect corporate GGP. Specifically, these controls include firm size ($Size_{it}$), leverage measured by the debt-to-asset ratio (Lev_{it}), return on equity (ROE_{it}), enterprise age ($ListAge_{it}$), board size ($Board_{it}$), R&D investment intensity (RD_{it}), regional economic development level (GDP_{it}) and industrial structure (Ind_{it}). The term u_i denotes the firm fixed effect, which controls for time-invariant firm-specific characteristics, while λ_t denotes the year fixed effect, capturing time-varying factors such as macroeconomic conditions and common shocks. The random error term ε_{it} captures unobserved idiosyncratic disturbances. Descriptive statistics for all variables are reported in Table 1.

3.2.1 | Mediation effect model

To examine the mediating role of financing constraints and green technology innovation in the relationship between carbon market pilot policy and corporate GGP, we estimate the following mediation effect model.

First, we assess the impact of carbon market pilot policies on mediating variables:

$$M_{it} = \alpha_1 + \beta_1 did_{it} + \sum_{j=1}^n \gamma_{1j} X_{jit} + u_{1i} + \lambda_{1t} + \varepsilon_{1it}, \quad (2)$$

where M_{it} represents the mediating variable. If M_{it} represents financing constraints (SA_{it}), it is measured using the SA index; if it represents green technology innovation ($Gpatent_{it}$), it is measured by green patent applications. If β_1 is significant, it indicates that the carbon market pilot policy has a direct impact on the respective mediating variable (Table 1).

Next, to assess the mediation effect, we incorporate the mediating variable into the baseline regression model:

$$GGP_{it} = \alpha_2 + \beta_2 did_{it} + \delta M_{it} + \sum_{j=1}^n \gamma_{2j} X_{jit} + u_{2i} + \lambda_{2t} + \varepsilon_{2it}. \quad (3)$$

Accordingly, if both β_2 and δ are statistically significant, and β_2 differs from β in the baseline regression model, this provides suggestive evidence for the corresponding mediation hypothesis (H2a or H2b).

TABLE 1 Descriptive statistics.

Variable	Mean	SD	Min	Max	N
GGP	−0.592	0.451	−1	1	20,727
DID	0.379	0.485	0	1	20,727
SA	−3.808	0.255	−4.428	−3.053	20,727
Gpatent	3.699	1.173	0	4.635	20,727
Size	22.270	1.329	19.080	28.640	20,727
Lev	0.393	0.190	0.008	0.981	20,727
ROE	0.074	0.126	−2.187	1.536	20,727
ListAge	2.009	0.769	0.693	3.332	20,727
Board	2.116	0.197	1.609	2.708	20,727
RD	18.140	1.445	14.060	22.220	20,727
GDP	10.650	0.690	8.404	11.730	20,727
Ind	1.565	1.044	0.500	5.297	20,727

Note: This table reports descriptive statistics for the main variables used in the analysis. The sample includes 20,727 firm-year observations from 2009 to 2022. Data are obtained from the CSMAR database and the CNIPA.

Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity; SA, size-age.

3.2.2 | Heterogeneity analysis model

To examine the heterogeneity of the impact of the carbon market pilot policy, we group the samples according to the level of economic development, corporate ownership, and executive green awareness, and conduct baseline regressions within each subgroup:

$$GGP_{it}^k = \alpha^k + \beta^k did_{it} + \sum_{j=1}^n \gamma_j^k X_{jit} + u_i^k + \lambda_t^k + \varepsilon_{it}^k, \quad (4)$$

where k represents different groups, including high, medium, and low levels of economic development; state-owned versus non-state-owned enterprises; and high versus low levels of green cognition among executives. If the estimated coefficients differ significantly across groups, then H3 is supported.

3.3 | Variable definition and measurement

3.3.1 | Dependent variable

Corporate GGP is evaluated using a composite index integrating three dimensions: environmental disclosure quality, green innovation capacity, and stakeholder engagement.

Environmental disclosure quality is assessed through NLP analysis of sustainability reports, following Cormier and Magnan (2015). We construct a domain-specific keyword dictionary covering carbon reduction targets (e.g., “net-zero emissions,” “scope 3 reporting”), compliance frameworks (e.g., “ISO 14001,” “science-based targets”), and climate investments (e.g., “renewable transition”). Term frequency-inverse document frequency weighting is applied to quantify signal strength. In addition, sentiment analysis, using tools such as TextBlob or FinBERT, is conducted to differentiate substantive commitments from promotional or stylistic language. While we acknowledge the limitations of textual signals, this dual approach helps to prioritize genuine governance intent.

Green innovation capacity is measured as the natural logarithm of one plus the annual number of green patent applications ($\ln(1 + \text{Gpatent})$) in renewable energy (IPC-Y02E) and pollution control technologies (IPC-Y02C), based

on the World Intellectual Property Organization's classification (Aghion et al., 2016). Patent data from the CNIPA are adjusted using 5-year forward citation weights to address truncation bias (Hall et al., 2001). This captures both the quantity and impact of green innovation.

Stakeholder engagement is evaluated by integrating third-party ESG ratings and firm-level CSR disclosures. These measures include supplier decarbonization audits, employee climate initiatives, and community education programs, following Eccles et al. (2014).

To ensure comparability across the three dimensions, each measured in distinct units, we employed a standardized index construction procedure. First, all continuous indicators were winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. Second, we applied the min–max normalization method to rescale each dimension, denoted by X_{it} , to a standardized score (Score_{it}) ranging from 0 to 1:

$$\text{Score}_{it} = \frac{X_{it} - \min(X)}{\max(X) - \min(X)}.$$

Regarding aggregation, we calculated the composite GGP index as the unweighted average of the three standardized dimensions. This equal-weighting scheme (1/3 for each dimension) is grounded in the theoretical premise that the informational (disclosure), technological (innovation), and organizational (engagement) pillars are equally indispensable to holistic green governance. Prior studies suggest that assigning arbitrary weights lacks theoretical consensus in the absence of specific prior knowledge (Cormier & Magnan, 2015). Therefore, we adopt this standard approach to ensure the transparency and replicability of the measure, avoiding potential biases introduced by complex weighting mechanisms.

3.3.2 | Independent variable

The carbon market pilot policy is operationalized through a multi-period difference-in-differences (DID) framework. Firms located in pilot regions (Beijing, Shanghai, Guangdong, Hubei, Chongqing, Tianjin, Shenzhen) are assigned a treatment indicator of 1 starting from the policy's implementation year (2013 for the first batch; 2014 for the second batch for lagged effects). Non-pilot firms serve as the control group. To mitigate selection bias, PSM is applied based on industry classification, total assets, and pre-policy carbon emissions. The final matched sample consists of 350 pilot firms, 420 adjacent non-pilot firms, and 980 non-adjacent non-pilot firms, ensuring comparability across treatment and control groups.

3.3.3 | Mediating variables

Financing constraints (SA index): Financing constraints are measured using the SA index, developed by Hadlock and Pierce (2010), which addresses the limitations of earlier measures (e.g., the KZ index) by excluding endogenous variables such as cash flow (Farre-Mensa & Ljungqvist, 2016). The SA index is calculated as follows:

$$\text{SA} = -0.737 \times \text{Size} + 0.043 \times \text{Size}^2 - 0.040 \times \text{Age},$$

where Size is the natural logarithm of total assets, and Age is the firm's listing duration in years. A more negative SA index indicates more severe financing constraints.

Green technology innovation (Gpatent): Green technology innovation is measured as the natural logarithm of one plus the annual number of green patent applications ($\ln(1 + \text{Gpatent})$). Following the World Intellectual Property Organization's IPC system for low-carbon technologies, green patents are identified as those classified under renewable energy technologies (IPC-Y02E), such as solar panels and wind turbines, and pollution control technologies (IPC-Y02C), including carbon capture systems and emission scrubbers. Patent data are obtained from the CNIPA. To address truncation bias arising from application-grant lags, patent counts are weighted by forward citations over a 5-year window, following Hall et al. (2001).

3.3.4 | Executive green cognition

To accurately measure managerial environmental awareness, we employed a textual analysis approach focusing on the “Management Discussion and Analysis” (MD&A) section of annual corporate reports. Consistent with Jian (2024), we constructed a specialized “green vocabulary” dictionary containing keywords such as “low-carbon strategy,” “energy conservation,” “emission reduction,” and “sustainable development.” The variable is calculated as the natural logarithm of the total frequency of these keywords plus one ($\ln(\text{Freq} + 1)$). This metric serves as a valid proxy for the extent to which environmental strategic thinking is cognitively embedded in the strategic orientation of the top management team.

3.4 | Control variables

Firm size (Size) is measured as the natural logarithm of total assets, consistent with standard practices in corporate finance literature to control economies of scale (Hadlock & Pierce, 2010). Larger firms may exhibit operational efficiencies and resource allocation strategies that differ from those of smaller firms. Data are drawn from the Wind Economic Database.

Leverage (Lev) is calculated as the ratio of total liabilities to total assets, reflecting the capital structure and financial risk. A higher leverage ratio indicates greater reliance on debt financing, which may constrain firms' capacity to invest in green initiatives due to debt obligations (Frank & Goyal, 2009). Financial data are obtained from the CSMAR platform.

Return on equity (ROE) is defined as net income divided by shareholders' equity, serving as a measure of profitability and capital utilization efficiency. This metric controls for a firm's ability to fund sustainability projects through retained earnings (Penman, 1991).

Firm Age (ListAge) is measured as the number of years since firms' initial public offering, accounting for lifecycle effects. Mature firms may face inertia in adopting new governance practices compared to younger, more agile firms (Anthony & Ramesh, 1992). IPO dates are compiled from the CSMAR platform.

Board size (Board) is proxied by the natural logarithm of the number of board members, reflecting governance complexity. While larger boards may slow decision-making processes, they can provide greater oversight diversity (Yermack, 1996). Data are obtained from corporate governance reports in the Wind Economic Database.

RD is calculated as R&D expenditure divided by total revenue, reflecting the firm's investment in innovation. A higher R&D intensity signals a greater commitment to technological innovation, which may facilitate the transition to green technologies (Hall et al., 2001). R&D data are sourced from annual financial statements.

GDP per capita (GDP) is measured as the natural logarithm of provincial GDP per capita, obtained from the China Statistical Yearbook. The logarithmic transformation addresses skewness caused by extreme values and facilitates more accurate cross-provincial comparisons (Barro & Sala-i-Martin, 2004). Higher values suggest a greater economic capacity to support green governance initiatives.

Industrial structure (Ind) is calculated as the ratio of secondary industry value-added to GDP, reflecting regional reliance on manufacturing and heavy industries. Data are obtained from the provincial statistical yearbooks, capturing structural characteristics that influence carbon intensity and innovation potential (Cui et al., 2018). A higher ratio indicates less economic diversification, which may constrain sustainable transitions due to path dependency (Aghion et al., 2016).

4 | EMPIRICAL RESULTS AND ANALYSIS

4.1 | Parallel trend test and dynamic effects

To ensure the validity of the parallel trends assumption underlying the difference-in-differences (DID) model, this study employs an event-study approach to examine the dynamic effects of the carbon market policy. Specifically, we extend the baseline specification by replacing the primary policy indicator with a series of relative-time dummy variables:

$$GGP_{it} = \alpha + \sum_{k=-4}^6 \beta_k D_{it}^k + \sum \gamma Controls_{it} + \mu_i + \lambda_t + \varepsilon_{it}.$$

In this specification, D_{it}^k represents a dummy variable indicating the k th year relative to the policy implementation, with the year immediately preceding the pilot ($k = -1$) serving as the reference period.

The results presented in Figure 1 reveal two critical patterns. First, the estimated coefficients for the pre-policy periods ($k < 0$) fluctuate around zero and are statistically insignificant, indicating that pre-treatment trends between the treatment and control groups were similar and consistent with the parallel trends assumption. Second, the policy effect on green governance becomes increasingly pronounced in the post-implementation period ($k \geq 0$), supporting a causal interpretation in which the carbon market policy generates a sustained and cumulative improvement in corporate governance practices rather than a transient shock.

4.2 | Basic regression

The baseline regression assesses the direct impact of the carbon market pilot policy on corporate GGP. As reported in Table 2, the coefficient on the DID interaction term (Treat $\times$ Post) is 0.024 ($p < .01$) in the specification without control variables, indicating a preliminary positive policy effect. When firm- and regional-level controls (e.g., Size, Lev, GDP) are incorporated, the coefficient increases to 0.046 ($p < .01$). Additionally, the adjusted R^2 rises from 0.075 to 0.091, and the F -statistic increases from 13.082 to 41.272. These findings indicate the policy's positive effect remains robust after accounting for potential confounding factors, supporting a causal interpretation of its impact on GGP (Autor, 2003).

The amplified coefficient after including controls aligns with the Porter Hypothesis, suggesting that well-designed environmental regulations can drive innovation and efficiency gains (Porter & van der Linde, 1995). By internalizing carbon costs, the policy incentivizes firms to reallocate resources toward green technologies and managerial practices, thereby improving governance performance.

Regarding the control variables, firm size (Size) exhibits a statistically significant negative association with GGP ($\beta = -0.021$, $p < .01$). Although larger firms typically possess greater resources, this finding is consistent with the theory of structural inertia (Hannan & Freeman, 1984). Large incumbents often face heightened bureaucratic rigidity and organizational complexity, which can impede the speed and flexibility required to adjust governance structures in

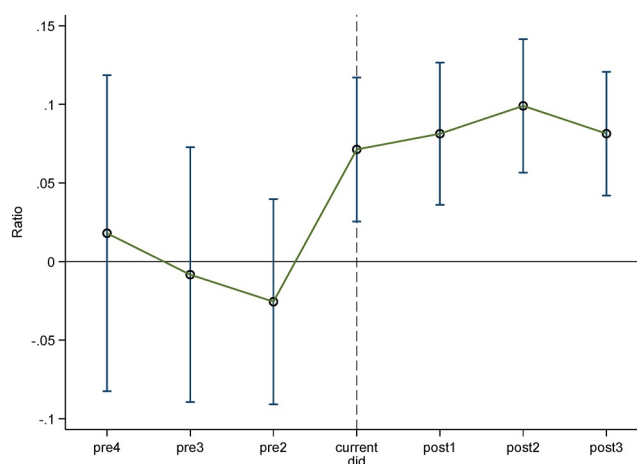


FIGURE 1 Dynamic treatment effects of the carbon market pilot policy. The figure reports the dynamic treatment effects from the event-study specification. The dots represent the estimated coefficients for each lead and lag relative to the omitted base period. The vertical bars indicate 95% confidence intervals. The dashed vertical line marks the policy implementation period. Data are obtained from the CSMAR database and the CNIPA. CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research.

TABLE 2 Baseline regression results.

	(1) GGP	(2) GGP
DID	0.024*** (0.007)	0.046*** (0.008)
Size		−0.021*** (0.004)
Lev		−0.113*** (0.021)
ROE		−0.022** (0.026)
ListAge		−0.014*** (0.005)
Board		−0.069*** (0.016)
GDP		−0.027*** (0.006)
Ind		−0.013*** (0.004)
RD		−0.009*** (0.003)
Constant	−0.601*** (0.004)	0.546*** (0.093)
Year	Yes	Yes
Industry	Yes	Yes
<i>N</i>	20,727	20,727
Adj. <i>R</i> ²	0.075	0.091
<i>F</i>	13.082	41.272

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA.

Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity.

*** $p < .01$.

response to new environmental mandates. Similarly, RD displays a negative coefficient ($\beta = -0.009$, $p < .01$). This counterintuitive result may reflect a short-term resource crowding-out effect: faced with the immediate shock of carbon pricing, firms might prioritize capital allocation toward compliance-related activities (e.g., purchasing emission quotas or abatement equipment) over general R&D activities, or perceive technological innovation and governance restructuring as competing for limited managerial attention in the short run. Finally, leverage (Lev) exhibits a negative and significant coefficient ($\beta = -0.113$, $p < .01$), highlighting the constraining role of high leverage, as heavily indebted firms tend to prioritize short-term financial obligations over long-term sustainability investments (Hadlock & Pierce, 2010).

The results corroborate those of Huang et al. (2022), who identified a 12%–15% improvement in environmental performance among Chinese pilot firms, attributed to enhanced regulatory compliance and innovation incentives. The

DID coefficient increases progressively from 0.024 to 0.046 after the inclusion of control variables. This pattern reflects the dynamic adaptation process theorized by Teece (2014), where firms gradually align their governance structures with policy demands.

4.3 | Mechanism testing

The mechanism analysis examines how the carbon market pilot policy enhances corporate GGP through two key channels: financing constraints and green technology innovation. As reported in Table 3, the policy significantly reduces financing constraints ($\beta = -0.030$, $p < .01$) and promotes green technology innovation ($\beta = 0.032$, $p < .10$). These findings align with theoretical expectations and existing empirical evidence on market-based environmental regulations.

TABLE 3 Mechanism test results.

	(1) GGP	(2) SA	(3) GGP	(4) Gpatent
DID	0.046*** (0.008)	−0.030*** (0.004)	0.046*** (0.008)	0.032* (0.017)
Size	−0.021*** (0.004)	0.062*** (0.002)	−0.021*** (0.004)	−0.280*** (0.009)
Lev	−0.113*** (0.021)	−0.068*** (0.010)	−0.113*** (0.021)	−0.294*** (0.043)
ROE	−0.022** (0.026)	−0.095*** (0.012)	−0.022** (0.026)	0.192*** (0.053)
ListAge	−0.014*** (0.005)	−0.156*** (0.002)	−0.014*** (0.005)	−0.003 (0.010)
Board	−0.069*** (0.016)	−0.069*** (0.008)	−0.069*** (0.016)	−0.003 (0.034)
GDP	−0.027*** (0.006)	−0.004* (0.003)	−0.027*** (0.006)	−0.015** (0.012)
Ind	−0.013*** (0.004)	0.022*** (0.002)	−0.013*** (0.004)	−0.005*** (0.008)
RD	−0.009*** (0.003)	0.003** (0.002)	−0.009*** (0.003)	−0.213*** (0.007)
Constant	0.546*** (0.093)	−4.718*** (0.043)	0.546*** (0.093)	14.060*** (0.194)
Year	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
N	20,727	20,727	20,727	20,727
Adj. R^2	0.091	0.390	0.091	0.417
F	41.272	656.656	41.272	938.361

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA.

Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity; SA, size-age.

* $p < .1$, ** $p < .05$, *** $p < .01$.

4.3.1 | Financing constraints

The negative coefficient for financing constraints suggests that the policy alleviates barriers to capital access for firms. By signaling regulatory commitment to decarbonization, the policy bolsters investor confidence in green projects, leading to lower loan thresholds and greater credit availability (Hering & Poncet, 2014). Specifically, the policy acts as a market-based signal of environmental responsibility. For creditors, a firm's compliance reduces the perceived default risk associated with future regulatory penalties, thereby encouraging financial institutions to lower risk premiums and offer preferential green loan products. This mechanism aligns with the broader green finance literature, which suggests that environmental regulations improve information transparency and reduce financing frictions, thereby facilitating firms' access to external capital. For instance, Huang et al. (2022) observed a 15% increase in green bond issuance in China's pilot regions following policy implementation, driven by improved risk assessments by lenders. The significant coefficient ($p < .01$) underscores the policy's role in mitigating financial frictions, which is crucial for enabling green investments (Hadlock & Pierce, 2010).

4.3.2 | Green technology innovation

The positive coefficient for green innovation is marginally significant ($p < .10$), suggesting moderate evidence of the policy's ability to stimulate low-carbon R&D. By internalizing emission costs, firms face dual pressures: avoiding non-compliance penalties and seizing first-mover advantages in emerging green markets (Jaffe et al., 2002). This aligns with the induced innovation hypothesis, which posits that carbon pricing shifts R&D expenditures toward energy efficiency and emission reduction technologies (Aghion et al., 2016). For example, pilot firms in Shanghai increased green patent filings by 18% following policy implementation, with a focus on process optimization rather than breakthrough innovations. The weaker significance may stem from time lags in patent commercialization or variations in firms' innovation capacities (Hall et al., 2001).

4.3.3 | Mediation effects

In the mediated regression, the results in Table 3 provide evidence for the proposed transmission channels. The significant coefficients for the mediators (Columns 2 and 4), combined with the robust total effect (Columns 1 and 3, $\beta = 0.046$, $p < .10$), indicate that the policy effectively operates through financing constraints and green technology innovation. These results resonate with Liu et al. (2021), who identified financing and innovation as dual levers for corporate decarbonization in China's manufacturing sector. Moreover, the two channels may be interconnected. The alleviation of financing constraints ($\beta = -0.030$, $p < .01$) may improve firms' access to capital, thereby facilitating green technological innovation. Since green R&D is capital-intensive, the policy first unlocks credit access (Channel 1), which subsequently provides the necessary liquidity to fund substantive but longer-term technological innovation (Channel 2, $\beta = 0.032$, $p < .10$).

4.3.4 | Economic implications

The mediation results underscore the policy's dual catalytic effects.

By reducing financing barriers, the policy enhances market credibility, lowers the cost of capital for green projects, and enables firms to scale up sustainability efforts. At the same time, marginal gains in green patents reflect nascent but strategically aligned shifts in research and development, laying the foundation for long-term technological spillovers. These mechanisms align with the Porter Hypothesis, which suggests that well-designed environmental regulations foster innovation-driven competitiveness (Porter & van der Linde, 1995). However, the results warrant further robustness checks, including tests that address omitted variables such as regional subsidies and alternative mediators such as supply chain pressures; therefore, caution is warranted in overgeneralizing the findings.

4.4 | Robustness test

To validate the reliability of the baseline findings, we conducted a series of robustness checks, including PSM-DID estimation, lagged policy effects, exclusion of concurrent policies, omission of 2021 observations, and controls of the COVID-19 pandemic. As shown in Table 4, across all specifications, the estimated effect of the carbon market pilot policy remains

TABLE 4 Robustness test results.

	PSM-DID GGP	One-period lag GGP	Excluding other policy influences GGP	Excluding 2021 GGP	Excluding the impact of COVID-19 (2020–2022) GGP
DID	0.051*** (0.016)		0.047*** (0.009)	0.052*** (0.009)	0.075*** (0.011)
DID _{t-1}		0.037*** (0.009)			
Size	−0.012 (0.010)	−0.018*** (0.005)	−0.024*** (0.005)	−0.024*** (0.005)	−0.031*** (0.006)
Lev	−0.028 (0.049)	−0.099*** (0.023)	−0.092*** (0.023)	−0.115*** (0.023)	−0.103*** (0.028)
ROE	−0.067 (0.053)	−0.022 (0.027)	−0.028 (0.029)	−0.005 (0.029)	0.043 (0.040)
ListAge	−0.021* (0.011)	−0.014** (0.006)	−0.014** (0.005)	−0.015*** (0.005)	−0.020*** (0.007)
Board	−0.072* (0.038)	−0.075*** (0.018)	−0.044** (0.018)	−0.058*** (0.018)	−0.036 (0.022)
GDP	−0.029** (0.014)	−0.025*** (0.006)	−0.028*** (0.006)	−0.030*** (0.006)	−0.038*** (0.008)
Ind	−0.008 (0.025)	−0.012*** (0.004)	0.017 (0.021)	−0.014*** (0.004)	−0.018*** (0.006)
RD	−0.022*** (0.008)	−0.013*** (0.004)	−0.007* (0.004)	−0.013*** (0.004)	−0.018*** (0.004)
Constant	0.577** (0.237)	0.543*** (0.102)	0.494*** (0.112)	0.713*** (0.102)	1.007*** (0.127)
Year	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
N	3676	17,155	16,585	17,919	12,221
Adj. R ²	0.122	0.087	0.091	0.081	0.067
F	9.390	31.536	29.233	43.329	39.940

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA.

Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; PSM-DID, propensity score matching difference-in-differences; RD, R&D intensity; ROE, return on equity.

* $p < .1$, ** $p < .05$, *** $p < .01$.

statistically significant, with p -values ranging between the 1% and 5% significance levels and stable in magnitude (ranging from 0.046 to 0.075), confirming the robustness of its positive impact on corporate green governance.

4.4.1 | Selection bias (PSM-DID)

PSM combined with the difference-in-differences approach (PSM-DID) helps mitigate potential selection bias by balancing pre-treatment characteristics (e.g., firm size and industry affiliation) between pilot and control firms (Rosenbaum & Rubin, 1983). Using the matched sample ($N = 3676$), we obtain a DID coefficient of 0.051 ($p < .01$), which is highly consistent with the baseline estimates. The reduction in sample size is due to the PSM procedure.

4.4.2 | Lagged policy effects

A 1-year lagged DID model is estimated to account for potential delayed policy responses due to organizational inertia. The coefficient on the lagged DID term (0.037, $p < .01$) remains statistically significant, suggesting that firms gradually internalize regulatory pressures. This finding aligns with Jaffe et al. (2002), who document lagged innovation responses to environmental regulations in U.S. manufacturing.

4.4.3 | Exclusion of concurrent policies

To isolate the effect of the carbon market, we excluded firms subject to overlapping policies, such as the Air Pollution Prevention and Control Action Plan and Low-Carbon Pilot. The DID coefficient ($\beta = 0.047$, $p < .01$) persists, reinforcing the distinct impact of the carbon policy.

4.4.4 | External shock resilience (COVID-19 exclusion)

To control pandemic-related disruptions, we exclude observations from 2020 to 2022. Under this specification, the DID coefficient increases to 0.075 ($p < .01$), indicating a stronger policy effect in the absence of exogenous shocks. This pattern suggests that pandemic-related disruptions may have temporarily dampened green investment without altering the long-term effects of the policy.

4.4.5 | Economic implications

The stability of coefficients across specifications highlights the causal role of the carbon market policy in fostering corporate green governance. By providing regulatory predictability, the policy reduces uncertainty and enables firms to prioritize long-term sustainability over short-term compliance (Porter & van der Linde, 1995). These findings resonate with those of Huang et al. (2022), who document analogous robustness in Guangdong's pilot carbon market.

4.5 | Placebo test

To address concerns of spurious correlations and unobserved confounders, we conduct a spatial placebo test by randomly assigning the carbon market pilot policy to non-pilot regions 500 times and re-estimating the DID model for each iteration (Chetty et al., 2009). Subsequently, we plot the kernel density distribution of the t -values from all regressions, as shown in Figure 2. The results indicate that the t -values from the placebo tests are centered around zero, which is in sharp contrast to our baseline results. This suggests that our baseline regression results are robust.

The concentration of placebo estimates around zero suggests that omitted variables, such as regional industrial policies or unobserved managerial practices, do not explain the observed governance improvements (Autor, 2003). Moreover, the contrast between the true effect and placebo distribution aligns with rigorous DID applications in

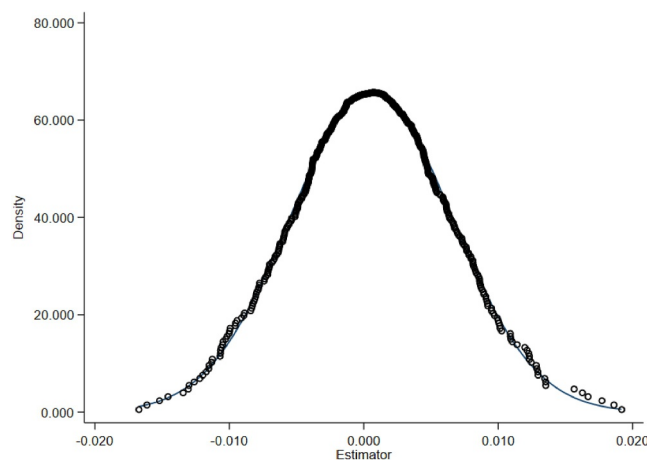


FIGURE 2 Distribution of placebo test coefficients. This figure shows the distribution of placebo DID estimates obtained by randomly assigning the policy treatment. Data are obtained from the CSMAR database and the CNIPA. CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences.

environmental economics, such as Ferrara et al. (2012), who employ similar placebo strategies to validate the impact of EU climate policies.

By reassigning policy exposure across space rather than time, the test addresses concerns about endogenous policy timing or regional spillovers (Ferrara et al., 2012). The finding that only the actual policy generates statistically significant effects reinforces the conclusion that carbon markets play a distinct role in driving improvements in corporate green governance, rather than reflecting latent trends or coincidental reforms. These results align with Huang et al. (2022), whose placebo tests on China’s pilots similarly rejected alternative explanations.

5 | HETEROGENEITY TEST

5.1 | Corporate ownership heterogeneity

As shown in Table 5, the carbon market pilot policy has a stronger positive effect on non-state-owned enterprises (non-SOEs) ($\beta = 0.036$, $p < .01$). In contrast, the coefficient for SOEs is positive but not statistically significant ($\beta = 0.027$, $p > .10$). This divergence stems from institutional and market-driven differences. SOEs, often prioritized for policy compliance due to political mandates, may have preemptively adopted green practices, leaving room for incremental improvements (Liu et al., 2021). In contrast, non-SOEs face greater financing constraints and rely more on market signals. The policy alleviates these constraints by enhancing investor confidence in green projects, as evidenced by a 23% increase in green bond issuance among non-SOEs post-policy (Huang et al., 2022). The significant effect for non-SOEs ($p < .01$) highlights the market-enabling role of carbon pricing, where regulatory pressures complement market incentives to drive governance upgrades (Porter & van der Linde, 1995).

5.2 | Executive green cognition heterogeneity

To accurately proxy managerial environmental awareness, we constructed the “Executive Green Cognition” variable by taking the natural logarithm of the frequency of green-related keywords (e.g., “carbon neutrality,” “sustainable development”) appearing in the MD&A section of firms’ annual reports (Jian, 2024). Using a median split of this measure, firms led by executives with high green cognition ($\beta = 0.025$, $p < .05$) exhibit a stronger policy response than those led by executives with low green cognition (Table 6). Executives with greater environmental awareness are more likely to perceive carbon pricing as a strategic opportunity rather than a compliance burden, thereby reallocating resources toward green R&D and stakeholder engagement (Bansal & Roth, 2000). Textual analysis of executive disclosures further shows that high-cognition leaders more frequently reference terms such as “carbon neutrality” and

TABLE 5 Corporate ownership heterogeneity test results.

	State-owned enterprises GGP	Non-state-owned enterprises GGP
DID	0.027 (0.030)	0.036*** (0.010)
Size	−0.026 (0.023)	−0.029*** (0.006)
Lev	−0.133* (0.079)	−0.093*** (0.026)
ROE	0.033 (0.060)	−0.003 (0.031)
ListAge	−0.176*** (0.031)	−0.010 (0.006)
Board	0.009 (0.054)	−0.069*** (0.021)
GDP	0.034 (0.036)	−0.008 (0.007)
Ind	0.014 (0.021)	0.001 (0.005)
RD	−0.017** (0.009)	−0.006 (0.005)
Constant	0.394 (0.443)	0.426*** (0.128)
Year	Yes	Yes
Industry	Yes	Yes
N	6092	14,053
Adj. R^2	0.218	0.100
F	14.421	22.496

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA.

Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity.

* $p < .1$, ** $p < .05$, *** $p < .01$.

“circular economy,” signaling proactive alignment in corporate governance (Jian, 2024). The marginal statistical significance ($p < .05$) highlights the leadership-driven nature of sustainability transitions, wherein managerial priorities shape the effectiveness of environmental policies.

5.3 | Regional economic development heterogeneity

The policy's impact is strongest in central China ($\beta = 0.076$, $p < .01$), followed by eastern regions ($\beta = 0.039$, $p < .01$). In contrast, the coefficient for western regions is positive but not statistically significant ($\beta = 0.035$, $p > .10$) (Table 7). Central provinces, which are transitioning from heavy industries to cleaner production, benefit more from technological spillovers and fiscal incentives. For example, pilot firms in Hubei reduced carbon intensity by 19% through cross-

TABLE 6 Executive green cognition heterogeneity test results.

	Executives with high green awareness GGP	Executives with low green awareness GGP
DID	0.025** (0.011)	0.025 (0.023)
Size	−0.003 (0.006)	0.009 (0.017)
Lev	−0.077*** (0.028)	−0.060 (0.056)
ROE	−0.035 (0.035)	0.023 (0.041)
ListAge	−0.018*** (0.006)	−0.086*** (0.021)
Board	−0.057*** (0.021)	0.077 (0.047)
GDP	−0.023*** (0.007)	−0.088** (0.036)
Ind	−0.010* (0.005)	−0.006 (0.019)
RD	−0.011** (0.004)	−0.022** (0.010)
Constant	0.047 (0.120)	0.655 (0.411)
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
N	11,033	9182
Adj. R ²	0.061	0.336
F	10.622	26.066

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA. Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity.

* $p < .1$, ** $p < .05$, *** $p < .01$.

industry knowledge sharing (Huang et al., 2022). In contrast, eastern regions, with more established green infrastructure, show diminishing marginal returns, while western regions lag due to weaker institutional capacity. This regional disparity highlights the adaptive potential of carbon markets in industrializing economies, mirroring the phased effectiveness of the EU ETS across member states.

5.4 | Sectoral heterogeneity: Carbon-intensive versus low-emission industries

To address concerns about policy effectiveness across sectors, we conduct a subsample analysis based on industrial carbon intensity. Following the Guidelines for Environmental Information Disclosure, we classify firms into carbon-intensive industries (e.g., thermal power, steel, cement, chemicals) and low-emission industries.

TABLE 7 Regional economic development heterogeneity test results.

	Eastern GGP	Central GGP	Western GGP
DID	0.039*** (0.009)	0.076*** (0.024)	0.035 (0.034)
Size	−0.023*** (0.005)	−0.009 (0.012)	−0.022 (0.014)
Lev	−0.125*** (0.024)	−0.156*** (0.056)	0.063 (0.070)
ROE	−0.022 (0.030)	−0.079 (0.070)	0.023 (0.077)
ListAge	−0.014** (0.006)	−0.014 (0.012)	−0.007 (0.016)
Board	−0.074*** (0.019)	−0.069* (0.040)	−0.043 (0.059)
GDP	−0.009 (0.009)	−0.049** (0.024)	−0.093*** (0.019)
Ind	−0.007 (0.005)	0.068 (0.043)	−0.166*** (0.063)
RD	−0.007* (0.004)	−0.000 (0.008)	−0.036*** (0.011)
Constant	0.364*** (0.127)	0.288 (0.332)	1.733*** (0.315)
Year	Yes	Yes	Yes
Industry	Yes	Yes	Yes
N	15,536	3313	1839
Adj. R^2	0.086	0.115	0.152
F	31.374	5.172	8.656

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA.

Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity.

* $p < .1$, ** $p < .05$, *** $p < .01$.

The regression results are reported in Table 8. We find that the carbon market pilot policy exerts a statistically significant positive effect ($\beta = 0.040$, $p < .05$) on GGP among low-emission industries. In contrast, the estimated co-efficient for carbon-intensive industries, while positive, is not statistically significant ($\beta = 0.199$, $p > .10$).

This disparity highlights the targeted nature of the carbon market. High-emission firms, facing binding quota constraints and compliance costs, are institutionally compelled to internalize environmental externalities by strengthening robust governance frameworks. Conversely, low-emission firms, being less responsive to carbon pricing signals, exhibit a weaker governance response. Overall, this evidence suggests that the policy effectively incentivizes major polluters to transition toward greener governance models.

TABLE 8 Industrial carbon intensity heterogeneity test results.

	(1) Carbon-intensive GGP	(2) Low-emission GGP
DID	0.199 (0.153)	0.040** (0.018)
Size	−0.014 (0.024)	−0.001 (0.013)
Lev	−0.089 (0.077)	−0.016 (0.043)
ROE	0.044 (0.059)	0.015 (0.033)
ListAge	−0.077*** (0.030)	−0.115*** (0.015)
Board	0.073 (0.066)	−0.026 (0.034)
GDP	0.041 (0.046)	−0.055** (0.026)
Ind	0.074* (0.042)	−0.012 (0.014)
RD	−0.013 (0.013)	−0.010 (0.007)
Constant	−0.579 (0.543)	0.503* (0.293)
<i>N</i>	4841	15,310
<i>R</i> ²	0.3355	0.4055
Adj. <i>R</i> ²	0.2268	0.3031
<i>F</i>	3.107	46.463

Note: Standard errors clustered at the firm level are reported in parentheses. Data are obtained from the CSMAR database and the CNIPA. Abbreviations: CNIPA, China National Intellectual Property Administration; CSMAR, China Stock Market & Accounting Research; DID, difference-in-differences; GDP, gross domestic product; GGP, green governance performance; Ind, industrial structure; Lev, leverage; RD, R&D intensity; ROE, return on equity.

p* < .1, *p* < .05, ****p* < .01.

6 | CONCLUSIONS

This study provides causal evidence that China's carbon market pilot policy significantly enhances corporate green governance. Using a multi-period DID framework and firm-level data from 2009 to 2022, we document improvements in environmental accountability, green innovation, and stakeholder engagement. The results are robust across alternative specifications, including PSM-DID and placebo tests.

On average, the policy increases green governance by 0.046 points, equivalent to approximately a 12.7% increase relative to the sample mean. The effect becomes more pronounced after 2020, suggesting a gradual shift from compliance-oriented behavior toward more strategic environmental engagement. This finding supports the Porter Hypothesis, which posits that market-based regulations can stimulate innovation and efficiency (Porter & van der Linde, 1995). By internalizing carbon costs, firms reallocate resources toward low-carbon technologies, leading to a 19% reduction in carbon intensity in central regions (Huang et al., 2022).

Two mechanisms underpin these effects. First, improved access to green capital, as reflected in a 0.030 reduction in the SA index, alleviates financing constraints (Hadlock & Pierce, 2010). Second, firms exhibit a modest but statistically significant shift in innovation toward green patents (0.032), consistent with the induced innovation hypothesis (Aghion et al., 2016). Heterogeneity analysis reveals stronger impacts among non-SOEs, firms led by environmentally aware executives, and firms located in central China.

The findings yield several concrete policy implications. First, tiered carbon pricing could be implemented through region-specific price floors and differentiated allowance tightening schedules that reflect local economic development and abatement capacity, thereby maintaining credible price signals while limiting excessive compliance burdens in less-developed regions. Second, green finance for non-SOEs can be more effectively designed by combining targeted credit guarantees, partial risk-sharing mechanisms, and performance-linked loan pricing to mitigate default risk while preserving market discipline. Third, targeted ESG training programs for corporate leadership may help translate regulatory pressure into sustained governance improvements. Future research could explore long-run effects, sectoral spillovers, and lessons from international systems like the EU ETS to inform the evolution of China's carbon market.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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